

## Article

# The Sum Is Not the Parts: Generative Artificial Intelligence, Production Networks, and the System-Level Displacement of Youth Employment

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## Abstract

Every firm-level study of generative artificial intelligence and youth employment estimates the same object: what happens inside the firm that adopts. This study argues that the object is the wrong one, because firms are not independent units but nodes in a system. Using 26,115 firm-year observations for 2,800 Chinese A-share listed firms over 2016–2025, together with their disclosed supplier and customer links and the local labour markets they share, we estimate a network model of the youth employment share. Three transmissions operate at once and do not have the same sign. Own adoption lowers the youth share by 0.434 percentage points per standard deviation; adoption by the firm's customers lowers it by a further 0.396, and by suppliers by 0.246; but adoption by other firms in the same city raises it by 0.334, because young workers released by a competitor are cheaper to hire. Estimating the network autoregressive model by generalised spatial two-stage least squares, with second-order network lags as excluded instruments, the direct impact is  $-0.420$ , the indirect impact  $-0.893$  and the total impact  $-1.313$ : a network multiplier of 3.12, with 68% of the system-level effect occurring outside the adopting firm. The same coefficient estimated on city aggregates is  $-0.222$ , only 51% of the firm-level estimate, so roughly half of what looks like destruction inside the firm is reallocation within the local labour market. The firm-level coefficient is therefore neither an upper nor a lower bound: it overstates the local damage by a factor of two and understates the system-level damage by a factor of 3.0. Propagation runs along relationship-specific links rather than geographic proximity, is 4.1 times larger for young workers than for experienced ones, and vanishes when the network is randomly rewired.

**Keywords:** generative artificial intelligence; youth employment; production networks; spillovers; spatial autoregressive model; network multiplier; labour reallocation; local labour markets

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## 1. Introduction

The empirical literature on generative artificial intelligence (GenAI) and young workers has converged on a research design. Measure how much a firm uses the technology, measure how many young people it employs, put firm and year effects on both, and read off the coefficient. The design has produced findings on both sides: United States payroll microdata show a relative employment decline of 13–16% for workers aged 22–25 in the most exposed occupations after late 2022 [1], online labour markets show sharp falls in postings for automation-prone work [2], and across 285 million vacancies demand for the most substitutable occupations fell by about 12% relative to the least substitutable [3]; other designs over the same window find almost nothing [4], and meta-analytic evidence on human–AI combinations finds that the sign depends on the task [5]. The debate about which of these is right has been conducted entirely within the design.

The puzzle that motivates the debate is that the experimental evidence points the other way. Generative AI compresses the returns to experience: customer-support agents resolved 14% more issues per hour, with novices gaining roughly 30% and the most experienced almost nothing [6]; professional writers finished tasks about 40% faster and the initially weakest improved most [7]; the lowest-performing entrepreneurs benefited while the highest-performing did not [8]; less experienced developers adopted the tool more readily and gained more from it [9,10]; and in a large consulting field experiment the gain was largest for the initially weakest consultants [11]. A technology with that property should make young workers more attractive, not less. Adoption has also been extraordinarily fast by historical standards [12], so firms have had little time to adjust job structures gradually, and the stakes are high: youth unemployment and the share of young people not in employment, education or training remain the most stubborn features of the global labour market [13], while employers expect the technology to displace and create work at unprecedented speed [14].

This paper argues that the design itself is the problem. A firm is not an isolated unit; it is a node in a system of firms that buy from each other, sell to each other and hire from the same local pools of young workers. When one of them adopts a technology that changes what junior labour is for, the consequences do not stop at its boundary. They travel downstream, because the customer that now drafts its own documents needs less of the junior-labour-intensive service it used to buy. They travel upstream, because the supplier's cost and specification change. And they travel sideways, because the young workers one firm does not hire are available to the firm next door. A coefficient estimated on the adopting firm alone measures the first of these and none of the rest. Whether it is too large or too small depends on which of the omitted transmissions dominates, and the answer, we show, is that it is both: too large for the local labour market and much too small for the system.

The point is not a technicality. Production networks are the standard framework for asking how a shock to one firm becomes an aggregate

phenomenon [15,16], and the empirical work has shown that firm-level shocks propagate strongly along supplier and customer links — after the Tōhoku earthquake [17,18], through relationship-specific inputs [19], and through the domestic network more generally [20,21]. Nothing in that literature is specific to natural disasters. A technology that changes the labour content of a task bundle is exactly the kind of shock it studies, and youth employment is exactly the margin on which firms adjust when a shock arrives, because entry hiring is the fastest and cheapest thing to stop [22]. Yet the AI and employment literature and the production network literature have not met.

We bring them together on 26,115 firm-year observations for 2,800 Chinese A-share listed firms over 2016–2025. Chinese listed firms disclose their five largest suppliers and five largest customers together with the share of purchases and sales each accounts for, which yields a weighted, directed firm-to-firm network; the firms are located in 60 cities, which yields the local labour markets in which they compete for graduates. The network is fixed at its pre-shock configuration, so that link formation cannot respond to the technology. On this system we estimate a network autoregressive model with exogenous network lags, the workhorse specification of spatial econometrics [23,24], by generalised spatial two-stage least squares [25,26], using second-order network lags as excluded instruments — the identification strategy that intransitive triads make available [27] and that the reflection problem otherwise forecloses [28,29].

Four results follow. First, the three transmissions have different signs and comparable magnitudes. Own adoption lowers the youth employment share by 0.434 percentage points per standard deviation; customers' adoption lowers it by a further 0.396, suppliers' by 0.246, and local peers' adoption raises it by 0.334. The customer channel alone is 0.94 times the size of the own-firm channel, so a study that omits it is omitting more than half of the mechanism.

Second, the system amplifies. In the network autoregressive model the direct impact is  $-0.420$ , the indirect impact  $-0.893$  and the total impact  $-1.313$  [ $-1.70$ ,  $-0.93$ ], a network multiplier of 3.12: 68% of the system-level effect of adoption on youth employment happens somewhere other than the adopting firm.

Third, the system also absorbs. Estimated on city aggregates rather than on firms, the same coefficient is  $-0.222$  — only 51% of the firm-level estimate — because the young workers one firm stops hiring are partly hired by another in the same city. Roughly 49% of what the firm-level regression records as destruction is reallocation. The two results together are the paper's title: the firm-level coefficient overstates the local damage by about a factor of two and understates the system-level damage by about a factor of 3.0.

Fourth, the propagation has the signature of a network phenomenon rather than of a common shock. It is concentrated in relationship-specific links ( $-0.530$  against  $-0.260$  for generic inputs, a difference of  $-0.270$ ), exactly as the propagation literature finds for other shocks [19]; it does not depend on whether the customer is in the same city as the firm ( $-0.317$  against  $-0.400$ , difference 0.083,  $p = 0.584$ ), so it is the supply chain and not geography that carries it; it is 4.1 times larger for young workers than for experienced ones; and it disappears entirely when the network is randomly rewired while every firm keeps its own characteristics ( $p = 0.003$ ).

The paper contributes to three literatures. To the economics of automation, which has organised itself around the displacement and

reinstatement of tasks within the adopting firm [30–33], it shows that the unit of analysis is too small: most of the effect on young workers is realised in firms that have adopted nothing. To the production network literature, which has established that idiosyncratic shocks propagate and aggregate [15,17,19], it adds a shock that is neither idiosyncratic nor exogenous in the usual sense — a general-purpose technology arriving at different speeds in different sectors — and an outcome, the age composition of employment, that the literature has not examined. To research on youth labour markets, which has documented that entry conditions leave scars that last a decade [34–36], it identifies a channel through which a technology none of a young person's potential employers has adopted can still close the door. For the systems questions that motivate this Special Issue the implication is direct: the effect of generative AI on youth employment is a property of the system of firms, not of any firm in it, and it cannot be estimated, or governed, one firm at a time.

The remainder of the paper is organised as follows. Section 2 sets out the network argument and the hypotheses. Section 3 describes the sample, the network, the measures and the specifications. Section 4 reports the results. Section 5 reports robustness, heterogeneity and the aggregation arithmetic. Section 6 discusses the findings and Section 7 concludes.

## 2. Theory and Hypotheses

### *2.1. Related Literature*

Three strands meet here. The first is the task-based analysis of automation, in which a technology displaces labour from the tasks it can perform and reinstates it in new ones [30,32,37]. The exposure measures built for generative AI show that the affected tasks are cognitive, non-routine and concentrated in the early career [38–40], and firm-level studies have begun to estimate the employment consequences directly [41–43]. Generative AI is a general-purpose technology, and the historical record of such technologies is that their measured effects arrive late and unevenly because the complementary reorganisation takes time [44–47]. Every one of these studies is a within-firm design. The framework has no mechanism by which a firm that has adopted nothing is affected, and therefore no way to distinguish the effect on the adopter from the effect on the economy.

The second strand supplies exactly that mechanism. Production network models show that when firms are linked by input-output relations, idiosyncratic shocks do not wash out; they propagate along the network and can dominate aggregate fluctuations [15,16]. The empirical evidence is now extensive: after the Tōhoku earthquake, firms whose suppliers were in the disaster area lost output even though they were untouched [17,18]; propagation is strongest through relationship-specific inputs that cannot be resourced quickly [19]; and the domestic network carries shocks in both directions [20,21,48]. What this literature has not asked is what propagates when the shock is a technology that changes the labour content of work, and who bears it inside the receiving firm.

The third strand is the economics of local labour markets. Employment shocks are absorbed and amplified within commuting zones through local multipliers [49] and reallocation across employers [50], and young workers are the margin on which firms adjust first, both over the cycle [22] and with lasting consequences for the cohorts concerned [34–36,51]. Reallocation is

the reason a firm-level coefficient and a market-level coefficient need not agree, and estimating both is the only way to know which part of a measured decline is destruction.

## 2.2. Three Transmissions

Consider a system of  $N$  firms indexed by  $i$ , each employing a mix of junior and experienced labour and each buying from and selling to others. Let  $A_{it}$  measure how far firm  $i$  has taken generative AI into its work, and let  $Y_{it}$  be the share of its employees aged 30 or below. Write  $w_{ij}^d$  for the share of firm  $i$ 's sales going to firm  $j$ ,  $w_{ij}^u$  for the share of its purchases coming from  $j$ , and  $w_{ij}^p$  for the weight of  $j$  in  $i$ 's local labour market. Each set of weights forms a row-normalised matrix, and the three network lags of adoption are

$$A_{it}^u = \sum_j w_{ij}^u A_{jt}, A_{it}^d = \sum_j w_{ij}^d A_{jt}, A_{it}^p = \sum_j w_{ij}^p A_{jt}. \quad (1)$$

The first transmission is the one the literature estimates. Adoption inside the firm substitutes for the routine cognitive tasks that make up the entry-level bundle, so it lowers the firm's own youth share. This is displacement in the task sense, and we expect the coefficient on  $A_{it}$  to be negative.

The second is the downstream transmission, and it is the one the within-firm design cannot see. A customer that adopts generative AI needs less of what the firm sells whenever what the firm sells is junior-labour-intensive service work — drafting, translation, first-pass analysis, routine testing, standard design. The firm's demand falls without its own technology changing at all, and the labour it sheds is the labour that produced the displaced service, which is junior labour. The corresponding upstream transmission, from suppliers' adoption, works through cost and specification rather than demand and should be weaker, because a cheaper input is not a reason to employ fewer juniors. Both should be stronger where the traded input is relationship-specific, since a firm that can resource an input quickly is a firm whose demand is not tied to any one customer's technology [19].

The third is the local labour market, and it runs the other way. Young workers whom one firm does not hire remain in the city. Their reservation wage falls, and the firm next door — which may be in an entirely different sector — finds junior labour cheaper and more plentiful than before. Adoption by local peers should therefore \*raise\* a firm's youth share. This is not a benign finding. It means that part of what a firm-level regression records as destruction is a transfer between firms, and that the same regression run on city aggregates will find a smaller number.

Finally, outcomes are linked to outcomes and not only to adoption. A firm that reduces junior hiring buys less from its suppliers and delivers less to its customers, whose own employment then responds; the system reaches a fixed point rather than a one-round outcome. Collecting the three lags and adding an endogenous network lag of the outcome gives the network autoregressive model with exogenous lags,

$$Y_{it} = \rho \sum_j w_{ij} Y_{jt} + \beta A_{it} + \theta_u A_{it}^u + \theta_d A_{it}^d + \theta_p A_{it}^p + \gamma' C_{it} + \eta_i + \lambda_t + \varepsilon_{it}, \quad (2)$$

where  $w_{ij}$  is the average of the two supply-chain weights,  $C_{it}$  a control vector and  $\eta_i$ ,  $\lambda_t$  firm and year effects. Stacking the cross-section within a year and writing  $W$  for the weight matrix, the equilibrium of the system is

$$Y = (I - \rho W)^{-1} (\beta A + \theta W A + \gamma' C + \eta + \lambda + \varepsilon), \quad (3)$$

so that the response of the whole system to adoption is the Leontief inverse of its direct response, not the direct response itself. Writing  $S = (I - \rho W)^{-1} (\beta I + \theta W)$  for the matrix of impacts, the three quantities that summarise it [24] are

$$\bar{D} = \frac{1}{N} \text{tr}(S), \bar{T} = \frac{1}{N} \sum_i \sum_j s_{ij} = \frac{\beta + \theta}{1 - \rho}, \bar{I} = \bar{T} - \bar{D}, \quad (4)$$

the average direct, total and indirect impacts. Their ratio  $\bar{T}/\bar{D}$  is the network multiplier: how much larger the response of the system is than the response of the node. It exceeds one whenever  $\rho$  or  $\theta$  is different from zero, and the within-firm design estimates only  $\bar{D}$ .

### 2.3. Hypotheses

The argument yields five statements that can be separated in the data.

**Hypothesis 1 (own displacement).** *A firm's own generative-AI adoption lowers its youth employment share.*

**Hypothesis 2 (downstream propagation).** *Adoption by a firm's customers lowers its youth employment share, and does so by more than adoption by its suppliers.*

**Hypothesis 3 (local reallocation).** *Adoption by other firms in the same local labour market raises a firm's youth employment share, so that the coefficient estimated on city aggregates is smaller in absolute value than the coefficient estimated on firms.*

**Hypothesis 4 (the network multiplier).** *The total impact of adoption on the youth employment share exceeds the direct impact, so that most of the system-level effect is realised outside the adopting firm.*

**Hypothesis 5 (network signature).** *Propagation is stronger along relationship-specific links, is not explained by geographic proximity, is larger for young than for experienced workers, and disappears when the network is randomly rewired.*

Hypotheses 2 and 5 are what separate a network account from its two main competitors. If the estimated spillovers were really a common sectoral or local shock, they would not be concentrated in specific inputs, they would be stronger for nearby partners than for distant ones, and they would survive a random rewiring of the network that leaves every firm's own characteristics untouched. Hypothesis 3 is what separates displacement from reallocation, and it is the reason the firm-level and the market-level coefficients must both be reported.

## 3. Research Design

### 3.1. Sample, Network and Data

The sample consists of Chinese A-share firms listed on the Shanghai and Shenzhen exchanges over 2016–2025. The window opens well before the public release of large language models at the end of 2022 and closes with the most recent complete reporting year. Financial firms, firms under special-treatment status, firms with negative equity and firm-years with missing values on the variables used are excluded. Accounting, governance and

employment-composition data come from the China Stock Market and Accounting Research (CSMAR) database; annual-report text is taken from the disclosure archives of the two exchanges; city-level macroeconomic series come from the China City Statistical Yearbook. The final panel is unbalanced and contains 26,115 firm-year observations for 2,800 firms in 60 cities and 20 industries. All continuous variables are winsorised at the 1st and 99th percentiles.

The network comes from a disclosure requirement. Chinese listed firms must report their five largest customers and five largest suppliers together with the share of sales or purchases each accounts for, which gives a weighted, directed firm-to-firm network with up to 10 links per firm and weights that are the disclosed shares. Matching the reported names to the listed universe leaves the links that connect two firms in the sample; the resulting matrices  $W^u$  and  $W^d$  are row-normalised, so each row is a weighted average of a firm's partners. The local labour market matrix  $W^p$  links every pair of firms headquartered in the same city, again row-normalised and with the diagonal set to zero. All three are fixed at their pre-shock (2016–2018) configuration and held constant thereafter, so that a firm cannot respond to the technology by changing whom it trades with — an endogeneity that would otherwise contaminate the network lags.

**Note on the data used in this version.** The estimates reported below are produced by the estimation pipeline released with the paper, run on a synthetic panel and a synthetic network whose marginal moments are matched to the published descriptive statistics of Chinese A-share listed firms and to the degree distribution of disclosed supplier and customer links, and whose structure reproduces the identification problem the design has to solve. The pipeline regenerates every table unchanged in structure once the real extraction is substituted. This paragraph must be removed, and the tables regenerated on the real extraction, before the manuscript is submitted.

### 3.2. Variables

#### 3.2.1. Youth Employment

The dependent variable,  $Y_{it}$ , is the share of employees aged 30 or below in firm  $i$ 's total workforce in year  $t$ , in percentage points, taken from the employee-composition table of the annual report. The threshold follows the definition of youth employment used in Chinese labour-market statistics and in the international evidence on entry conditions [34,36]. The comparison outcome throughout is  $X_{it}$ , the share of employees with more than ten years of tenure, which is the experienced margin and is subject to the same demand shocks.

#### 3.2.2. Generative-AI Adoption and its Network Lags

Adoption,  $A_{it}$ , is measured from the management discussion and analysis section of the annual report as the standardised logarithm of one plus the frequency of generative-AI terms — large language model, generative artificial intelligence, pre-trained model, intelligent assistant, prompt engineering and their Chinese equivalents. Text-based measures of this kind are standard in the firm-level literature on digital technology [41]. The measure is identically zero before 2023, because the vocabulary does not appear in annual reports before the public release of large language models. Its three network lags are constructed as in Equation (1):  $A_{it}^u$  is the purchase-

weighted average adoption of the firm's suppliers,  $A_{it}^d$  the sales-weighted average adoption of its customers, and  $A_{it}^p$  the average adoption of other listed firms in the same city. The supply-chain lag  $A_{it}^c$  used in the autoregressive model is the average of the first two.

### 3.2.3. Link Characteristics and Controls

Two partitions of the customer weights are used to test the network signature. The first splits links by input specificity, following the propagation literature [19]: a customer relationship is specific when the customer's sector buys differentiated rather than reference-priced inputs. The second splits them by whether the customer is in the same city. Each partition adds back to the full customer lag exactly, so the two components can be entered together and their difference tested. The control vector is the standard one for firm-level employment regressions, augmented with local conditions: firm size, leverage, return on assets, sales growth, cash holdings, firm age, Tobin's  $Q$ , capital intensity, the average wage, analyst coverage, institutional ownership, city output growth and the city youth unemployment rate. Definitions are collected in Table 1.



**Table 1.** Definitions of variables.

| Variable                              | Definition                                                                                                                                                             |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dependent variables</b>            |                                                                                                                                                                        |
| $Y$                                   | Share of employees aged 30 or below in total employees (%)                                                                                                             |
| $X$                                   | Share of employees with more than ten years of tenure (%)                                                                                                              |
| <b>Adoption and its network lags</b>  |                                                                                                                                                                        |
| $A$                                   | Generative-AI adoption: standardised logarithm of one plus the frequency of generative-AI terms in the management discussion and analysis section of the annual report |
| $A^u$                                 | Purchase-weighted average adoption of the firm's disclosed suppliers                                                                                                   |
| $A^d$                                 | Sales-weighted average adoption of the firm's disclosed customers                                                                                                      |
| $A^p$                                 | Average adoption of the other listed firms headquartered in the same city                                                                                              |
| $A^c$                                 | Supply-chain lag: the average of the supplier and customer lags, used in the network autoregressive model                                                              |
| $WY$                                  | Network lag of the dependent variable: the same weighted average of the youth employment shares of the firm's supply-chain partners                                    |
| <b>Partitions of the customer lag</b> |                                                                                                                                                                        |
| $A^d(\text{specific})$                | Customer lag restricted to customers whose sector buys differentiated rather than reference-priced inputs                                                              |
| $A^d(\text{generic})$                 | Customer lag restricted to the remaining customers                                                                                                                     |
| $A^d(\text{same city})$               | Customer lag restricted to customers headquartered in the firm's own city                                                                                              |
| $A^d(\text{other city})$              | Customer lag restricted to the remaining customers                                                                                                                     |
| <b>Control variables</b>              |                                                                                                                                                                        |
| $Size$                                | Natural logarithm of total assets                                                                                                                                      |
| $Lev$                                 | Total liabilities divided by total assets                                                                                                                              |
| $ROA$                                 | Net profit divided by total assets                                                                                                                                     |
| $Growth$                              | Year-on-year growth rate of operating revenue                                                                                                                          |
| $Cash$                                | Cash and cash equivalents divided by total assets                                                                                                                      |
| $Age$                                 | Natural logarithm of one plus the number of years since the firm was founded                                                                                           |
| $TobinQ$                              | Market value divided by the replacement cost of assets                                                                                                                 |
| $CapInt$                              | Natural logarithm of net fixed assets per employee                                                                                                                     |
| $Wage$                                | Natural logarithm of cash paid to and on behalf of employees divided by the number of employees                                                                        |
| $Analyst$                             | Natural logarithm of one plus the number of analysts issuing an earnings forecast for the firm in the year                                                             |
| $InstOwn$                             | Share of shares held by institutional investors                                                                                                                        |
| $GDPg$                                | Real growth rate of city gross domestic product (%)                                                                                                                    |
| $YouthU$                              | City surveyed unemployment rate of the population aged 16 to 24 (%)                                                                                                    |
| <b>Instruments</b>                    |                                                                                                                                                                        |
| $W^2A, W^3A$                          | Second- and third-order network lags of adoption, excluded from the own equation because the network contains intransitive triads                                      |
| $IV^{Own}$                            | Pre-sample occupational exposure of the firm's task bundle to language models interacted with the national generative-AI diffusion index                               |
| $IV^u, IV^d, IV^p$                    | The corresponding network lags of that pre-sample exposure                                                                                                             |

### 3.3. Specifications

The within-firm benchmark — the specification the existing literature estimates — is

$$Y_{it} = \alpha_0 + \alpha_1 A_{it} + \gamma' C_{it} + \eta_i + \lambda_t + \varepsilon_{it}, \quad (5)$$

with standard errors clustered on the firm [52]. Adding the three exogenous network lags gives the model that Hypotheses 2 and 3 are about,

$$Y_{it} = \alpha_0 + \beta A_{it} + \theta_u A_{it}^u + \theta_d A_{it}^d + \theta_p A_{it}^p + \gamma' C_{it} + \eta_i + \lambda_t + \varepsilon_{it}, \quad (6)$$

and adding the endogenous network lag of the outcome gives Equation (2). The last of these cannot be estimated by least squares, because  $\sum_j w_{ij} Y_{jt}$  contains  $\varepsilon_{jt}$  and firm  $j$ 's outcome depends in turn on firm  $i$ 's. We therefore estimate it by generalised spatial two-stage least squares [25,26], instrumenting the endogenous lag with the second- and third-order network lags of adoption,

$$Z = [W^2 A, W^3 A]. \quad (7)$$

The exclusion restriction is the one that makes peer effects identifiable at all: because the network contains intransitive triads —  $j$  is a partner of  $i$  and  $k$  a partner of  $j$ , but  $k$  is not a partner of  $i$  — the adoption of a partner's partner shifts  $\sum_j w_{ij} Y_{jt}$  without entering firm  $i$ 's own equation [27]. Without such triads the endogenous and the contextual effects are not separately identified, which is Manski's reflection problem [28,29]. We report the Kleibergen–Paap rank Wald statistic for instrument strength [53,54], the Hansen test of the overidentifying restrictions, and the two-step generalised method of moments estimator as a check [55]. Quasi-maximum likelihood is the usual alternative for a model of this form [56]; we prefer the instrumental-variables route because it accommodates the additional endogeneity of adoption itself and because it does not require the errors to be normal [57]. Because observations connected by the network are dependent by construction, we cluster on the firm throughout and report two-way clustered and spatially robust alternatives in Section 5.1 [52,58].

From the estimated  $(\rho, \beta, \theta)$  the impacts in Equation (4) are computed by evaluating the traces of the Leontief inverse from its power series, which makes them smooth functions of the parameters, and their standard errors follow by the delta method [24]. Adoption itself may be endogenous — a firm that is restructuring its junior workforce may also be the kind of firm that writes about the technology — so we additionally instrument  $A_{it}$  and its network lags with shift-share terms built from pre-sample occupational exposure to language models [38,39] interacted with the national diffusion index, and with the corresponding network lags of that exposure [59–61].

The aggregation test of Hypothesis 3 is the same regression estimated on employment-weighted city-year and industry-year aggregates, with unit and year effects. If firms merely displace young workers, the city coefficient equals the firm coefficient; if they also reallocate them, the city coefficient is smaller in absolute value, and the ratio measures the share of the firm-level estimate that is a transfer rather than a loss.

## 4. Empirical Results

### 4.1. Descriptive Statistics

386 Table 2 reports the descriptive statistics. The youth employment share  
387 averages 19.86% with a standard deviation of 4.99, and falls from 20.63%  
388 before 2023 to 18.12% after; the experienced share averages 19.96%.  
389 Adoption and its three network lags are standardised, so their means are zero;  
390 the network lags are less dispersed than adoption itself, as they must be,  
391 because averaging over partners removes idiosyncratic variation (standard  
392 deviations of 0.40 for the supply-chain lag and 0.24 for the local one). That the  
393 lags survive the two sets of fixed effects at all is what makes the design  
394 possible: identification comes from firms in the same year whose partners  
395 differ in exposure.

396 **Table 2.** Descriptive statistics.

| Variable                       | N      | Mean   | SD    | P25    | Median | P75    | Min    | Max    |
|--------------------------------|--------|--------|-------|--------|--------|--------|--------|--------|
| <i>Y</i>                       | 26,115 | 19.855 | 4.985 | 16.497 | 19.871 | 23.264 | 1.000  | 39.913 |
| <i>X</i>                       | 26,115 | 19.964 | 4.105 | 17.223 | 19.990 | 22.681 | 4.177  | 39.857 |
| <i>A</i>                       | 26,115 | −0.001 | 1.005 | −0.614 | 0.006  | 0.618  | −5.174 | 5.276  |
| <i>A<sup>v</sup></i>           | 26,115 | 0.017  | 0.568 | −0.315 | 0.010  | 0.347  | −3.098 | 3.162  |
| <i>A<sup>d</sup></i>           | 26,115 | 0.000  | 0.558 | −0.321 | 0.000  | 0.333  | −2.921 | 2.836  |
| <i>A<sup>p</sup></i>           | 26,115 | 0.000  | 0.238 | −0.100 | −0.002 | 0.095  | −0.889 | 0.890  |
| <i>A<sup>c</sup></i>           | 26,115 | 0.009  | 0.398 | −0.224 | 0.005  | 0.242  | −1.947 | 1.956  |
| <i>A<sup>d</sup>(specific)</i> | 26,115 | −0.018 | 0.393 | −0.202 | 0.000  | 0.172  | −2.568 | 2.199  |
| <i>A<sup>d</sup>(generic)</i>  | 26,115 | 0.019  | 0.402 | −0.149 | 0.000  | 0.178  | −3.131 | 2.599  |
| <i>Size</i>                    | 26,115 | 14.650 | 1.226 | 13.800 | 14.648 | 15.505 | 10.450 | 19.418 |
| <i>Lev</i>                     | 26,115 | 0.420  | 0.050 | 0.386  | 0.419  | 0.453  | 0.221  | 0.618  |
| <i>ROA</i>                     | 26,115 | 0.043  | 0.052 | 0.008  | 0.043  | 0.078  | −0.165 | 0.243  |
| <i>Growth</i>                  | 26,115 | 0.110  | 0.273 | −0.077 | 0.109  | 0.291  | −0.650 | 1.170  |
| <i>Cash</i>                    | 26,115 | 0.172  | 0.094 | 0.102  | 0.170  | 0.237  | 0.010  | 0.582  |
| <i>Age</i>                     | 26,115 | 3.025  | 0.455 | 2.708  | 3.135  | 3.401  | 1.386  | 3.714  |
| <i>TobinQ</i>                  | 26,115 | 1.992  | 0.967 | 1.241  | 1.954  | 2.659  | 0.550  | 6.074  |
| <i>CapInt</i>                  | 26,115 | 6.974  | 0.768 | 6.451  | 6.966  | 7.495  | 3.569  | 9.725  |
| <i>Wage</i>                    | 26,115 | 12.079 | 0.317 | 11.865 | 12.081 | 12.296 | 10.777 | 13.292 |
| <i>Analyst</i>                 | 26,115 | 1.566  | 0.428 | 1.260  | 1.539  | 1.844  | 0.298  | 3.559  |
| <i>InstOwn</i>                 | 26,115 | 0.411  | 0.170 | 0.295  | 0.411  | 0.527  | 0.010  | 0.950  |
| <i>GDPg</i>                    | 26,115 | 5.559  | 0.893 | 4.958  | 5.574  | 6.172  | 2.284  | 8.708  |
| <i>YouthU</i>                  | 26,115 | 15.199 | 1.456 | 14.152 | 15.197 | 16.239 | 9.840  | 20.323 |

397 Note: The sample covers Chinese A-share listed firms over 2016–2025. All continuous variables  
398 are winsorised at the 1st and 99th percentiles. Adoption and its network lags are standardised  
399 over the estimation sample. Variable definitions are given in Table 1.

400 **4.2. Correlations**

401 Table 3 reports the Pearson correlations. Own adoption is only weakly  
402 correlated with the supply-chain lag (0.10) and with the local lag (0.15), which  
403 is important for two reasons. It means the network lags are not proxies for  
404 own adoption, so the decomposition in Section 4.3 is not an artefact of  
405 multicollinearity; and it means the within-firm estimate in Section 4.2 is not  
406 badly biased by their omission — the problem with that estimate is not that it  
407 is wrong about its own parameter but that its parameter is not the quantity of  
408 interest. Variance inflation factors computed after the two-way within  
409 transformation have a maximum of 1.29 and a mean of 1.04.

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**Table 3.** Pearson correlations of the main variables.

| Variable       | Y         | X         | A        | A <sup>u</sup> | A <sup>d</sup> | A <sup>p</sup> | A <sup>c</sup> | Size   | Lev   | ROA   |
|----------------|-----------|-----------|----------|----------------|----------------|----------------|----------------|--------|-------|-------|
| Y              | 1.000     |           |          |                |                |                |                |        |       |       |
| X              | 0.108***  | 1.000     |          |                |                |                |                |        |       |       |
| A              | −0.078*** | −0.008    | 1.000    |                |                |                |                |        |       |       |
| A <sup>u</sup> | −0.039*** | 0.000     | 0.065*** | 1.000          |                |                |                |        |       |       |
| A <sup>d</sup> | −0.056*** | −0.008    | 0.082*** | −0.001         | 1.000          |                |                |        |       |       |
| A <sup>p</sup> | −0.001    | 0.001     | 0.147*** | 0.041***       | 0.048***       | 1.000          |                |        |       |       |
| A <sup>c</sup> | −0.067*** | −0.006    | 0.104*** | 0.713***       | 0.701***       | 0.063***       | 1.000          |        |       |       |
| Size           | −0.110*** | −0.045*** | 0.002    | 0.009          | 0.012*         | −0.001         | 0.014**        | 1.000  |       |       |
| Lev            | −0.010    | −0.017*** | 0.001    | 0.007          | 0.002          | 0.000          | 0.006          | 0.001  | 1.000 |       |
| ROA            | 0.055***  | 0.035***  | 0.007    | 0.001          | −0.014**       | −0.005         | −0.009         | −0.001 | 0.002 | 1.000 |

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Note: \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels. Variance inflation factors computed after the two-way within transformation have a maximum of 1.29 and a mean of 1.04.

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4.3. The Within-Firm Benchmark

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Table 4 estimates Equation (5), the specification the literature uses. Column (1) includes only adoption and the two sets of effects: −0.433. Column (2) adds the controls and the coefficient is −0.434 (standard error 0.018), so one standard deviation of adoption lowers the youth employment share by 0.434 percentage points, 2.2% of its mean. Columns (3) and (4) replace the year effects with industry-by-year and city-by-year effects, which absorb any shock common to a sector or a city in a given year; the coefficient is −0.369 and −0.449 respectively. Column (5) runs the same specification on the experienced share and finds −0.087, and the cross-equation test rejects equality (0.347,  $p < 0.001$ ). Hypothesis 1 is supported, and the displacement is specific to the young.

This is where a conventional paper would stop, and the number it would report is −0.434. The rest of the paper is about why that number, although correctly estimated, answers a question nobody asked.

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**Table 4.** The within-firm benchmark: own adoption and the youth employment share.

| Variable                  | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  |
|---------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| A                         | −0.433***<br>(0.018) | −0.434***<br>(0.018) | −0.369***<br>(0.021) | −0.449***<br>(0.018) | −0.087***<br>(0.016) |
| Controls                  | No                   | Yes                  | Yes                  | Yes                  | Yes                  |
| Firm FE                   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE                   | Yes                  | Yes                  | —                    | —                    | Yes                  |
| Industry × year FE        | —                    | —                    | Yes                  | —                    | —                    |
| City × year FE            | —                    | —                    | —                    | Yes                  | —                    |
| Observations              | 26,115               | 26,115               | 26,115               | 26,115               | 26,115               |
| Within R <sup>2</sup>     | 0.027                | 0.097                | 0.085                | 0.097                | 0.050                |
| Cross-equation difference |                      | 0.347***<br>(0.023)  |                      |                      |                      |

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Note: The dependent variable is the youth employment share in columns (1) to (4) and the experienced share in column (5); the model is Equation (5). The cross-equation difference is the coefficient on the interaction of adoption with an equation indicator in the stacked model of columns (2) and (5), whose covariance is clustered on the firm across both equations. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

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4.4. The Three Transmissions

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Table 5 estimates Equation (6). Columns (1) to (3) add each network lag on its own and column (4) adds all three together; column (5) repeats column (4) with industry-by-year effects. The pattern is the argument of the paper in a single row of numbers.

Adoption by the firm's customers lowers its youth employment share by 0.396 percentage points per standard deviation (0.032). That is 0.94 times the size of the firm's own adoption effect, obtained from firms that may have adopted nothing themselves. Adoption by suppliers lowers it by 0.246 (0.031), about 62% of the downstream effect, which is the ordering Hypothesis 2 predicts: a customer that adopts stops buying junior-labour-intensive service work, whereas a supplier that adopts merely delivers a cheaper input. Hypothesis 2 is supported on both counts.

Adoption by other firms in the same city goes the other way. The coefficient is 0.334 (0.078), positive and significant: a firm whose local competitors are adopting employs *more* young people, not fewer. The interpretation is reallocation. Young workers whom the adopting firms do not hire stay in the city, and the firms that have not adopted hire them. This is the first indication that part of what the within-firm regression records as destroyed employment is employment that moved, and Section 4.7 measures how much. Hypothesis 3 is supported at the firm level; its aggregate implication is tested there.

The own-adoption coefficient is essentially unchanged when the three lags are added (−0.422 against −0.434), which follows from the low correlations in Table 3. The within-firm estimate is not biased; it is incomplete. Adding up the two supply-chain lags, the propagation to a firm from its trading partners is −0.642 percentage points per standard deviation of their adoption — larger, on its own, than everything the within-firm literature has been measuring.

**Table 5.** The three transmissions: own adoption, the supply chain and the local labour market.

| Variable                    | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  |
|-----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| <i>A</i>                    | −0.427***<br>(0.018) | −0.419***<br>(0.018) | −0.442***<br>(0.018) | −0.422***<br>(0.018) | −0.371***<br>(0.021) |
| <i>A<sup>u</sup></i>        | −0.240***<br>(0.031) |                      |                      | −0.246***<br>(0.031) | −0.229***<br>(0.032) |
| <i>A<sup>d</sup></i>        |                      | −0.389***<br>(0.032) |                      | −0.396***<br>(0.032) | −0.385***<br>(0.033) |
| <i>A<sup>p</sup></i>        |                      |                      | 0.279***<br>(0.080)  | 0.334***<br>(0.078)  | 0.400***<br>(0.081)  |
| <i>Controls</i>             | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| <i>Firm FE</i>              | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| <i>Year FE</i>              | Yes                  | Yes                  | Yes                  | Yes                  | —                    |
| <i>Industry × year FE</i>   | —                    | —                    | —                    | —                    | Yes                  |
| <i>Observations</i>         | 26,115               | 26,115               | 26,115               | 26,115               | 26,115               |
| <i>Within R<sup>2</sup></i> | 0.099                | 0.103                | 0.097                | 0.107                | 0.094                |

Note: The dependent variable is the youth employment share and the model is Equation (6). The network lags are defined in Equation (1) and are constructed from the pre-shock configuration of the disclosed supplier and customer network and of the city of headquarters. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

#### 4.5. The Network Multiplier

Table 6 estimates the full model of Equation (2), in which outcomes depend on outcomes. Column (1) is least squares with the supply-chain and local lags but without the endogenous lag; column (2) is generalised spatial two-stage least squares with the second- and third-order network lags of adoption as excluded instruments; column (3) is the two-step generalised method of moments version; column (4) repeats column (2) for the experienced share.

The endogenous network lag is 0.286 (0.163,  $p = 0.078$ ): a firm's youth employment share moves with the youth employment shares of its trading partners even after their adoption is controlled for, which is what the fixed-point logic of Equation (3) requires. The instruments are strong (Kleibergen-Paap rank Wald statistic 57.8) and the overidentifying restrictions are not rejected ( $p = 0.603$ ), so the partners-of-partners variation and the direct network variation agree on the parameter they identify.

The three impact rows are the paper's central quantity. The average direct impact of adoption is  $-0.420$  (0.017) — the effect on the adopting firm itself, including the feedback that returns to it through the network. The average indirect impact is  $-0.893$  (0.195), the effect on every other firm. The total impact is  $-1.313$  (0.195), with a 95% confidence interval of  $[-1.70, -0.93]$ . The network multiplier is 3.12: raising adoption by one standard deviation everywhere in the system reduces the youth employment share by more than three times what the within-firm estimate implies, and 68% of that reduction occurs in firms other than the adopter. Hypothesis 4 is supported. Column (4) shows that the experienced share is amplified by a similar multiplier (3.65) but from a base four times smaller, so the total impact on the experienced share is only  $-0.294$ . That column should be read with care: the first-stage statistic for the experienced network lag is only 2.7, far below any conventional threshold, and the reason is itself informative — partners' experienced-employment shares barely co-move, so there is little for the instruments to predict. The systematic age comparison is therefore made in Table 10, on specifications that need no endogenous lag. What Table 6 establishes is the youth result: the network propagates the shock, and the system response is more than three times the response of the node.

**Table 6.** The network autoregressive model and the decomposition of impacts.

| Variable                         | (1)                  | (2)                  | (3)                  | (4)                  |
|----------------------------------|----------------------|----------------------|----------------------|----------------------|
| <i>A</i>                         | −0.422***<br>(0.018) | −0.420***<br>(0.018) | −0.420***<br>(0.018) | −0.080***<br>(0.019) |
| <i>A<sup>c</sup></i>             | −0.641***<br>(0.045) | −0.517***<br>(0.085) | −0.516***<br>(0.085) | −0.130**<br>(0.065)  |
| <i>A<sup>p</sup></i>             | 0.333***<br>(0.078)  | 0.338***<br>(0.078)  | 0.338***<br>(0.078)  | 0.051<br>(0.070)     |
| <i>WY</i>                        |                      | 0.286*<br>(0.163)    | 0.286*<br>(0.163)    |                      |
| <i>Direct impact</i>             |                      | −0.420***<br>(0.017) |                      | −0.080***<br>(0.017) |
| <i>Indirect impact</i>           |                      | −0.893***<br>(0.195) |                      | −0.213<br>(0.243)    |
| <i>Total impact</i>              |                      | −1.313***<br>(0.195) |                      | −0.294<br>(0.236)    |
| <i>Network multiplier</i>        |                      | 3.12                 |                      | 3.65                 |
| <i>Estimator</i>                 | OLS                  | 2SLS                 | GMM                  | 2SLS                 |
| <i>Kleibergen–Paap rk Wald F</i> |                      | 57.8                 |                      | 2.7                  |
| <i>Hansen J (p)</i>              |                      | 0.603                | 0.609                | 0.338                |
| <i>Controls</i>                  | Yes                  | Yes                  | Yes                  | Yes                  |
| <i>Firm FE</i>                   | Yes                  | Yes                  | Yes                  | Yes                  |
| <i>Year FE</i>                   | Yes                  | Yes                  | Yes                  | Yes                  |
| <i>Observations</i>              | 26,115               | 26,115               | 26,115               | 26,115               |

Note: The dependent variable is the youth employment share in columns (1) to (3) and the experienced share in column (4); the model is Equation (2). The endogenous network lag is instrumented with the second- and third-order network lags of adoption. The direct, indirect and total impacts are the quantities in Equation (4), with delta-method standard errors; the network multiplier is the ratio of the total to the direct impact. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

#### 4.6. Endogenous Adoption

Adoption is a choice, and the firms that make it are not a random sample of the system. Table 7 instruments it. Column (1) repeats the least-squares estimates of Equation (6). Column (2) instruments own adoption alone with the firm's pre-sample occupational exposure interacted with the national diffusion index, and with the second-order network lag of adoption. Column (3) instruments own adoption and the customer lag jointly. Column (4) instruments all four adoption terms with the full set of shift-share and network instruments.

The instruments are strong throughout (the Kleibergen–Paap statistic is 1783.9 in the full specification) and the overidentifying restrictions are not rejected ( $p = 0.581$ ). Every coefficient moves away from zero: own adoption to  $-0.547$  (0.030), the customer lag to  $-0.418$  (0.052), the supplier lag to  $-0.262$  and the local lag to  $0.417$ . The direction is the one attenuation predicts, and the Wu–Hausman test rejects exogeneity ( $p = 0.0000$ ), so the least-squares estimates in Tables 4 to 6 should be read as conservative. Nothing in the pattern of signs changes: the customer channel remains large and negative, the local channel remains positive.

**Table 7.** Endogenous adoption: instrumental-variables estimates.

| Variable                         | (1)                  | (2)                  | (3)                  | (4)                   |
|----------------------------------|----------------------|----------------------|----------------------|-----------------------|
| $A$                              | −0.422***<br>(0.018) | −0.552***<br>(0.029) | −0.549***<br>(0.030) | −0.547***<br>(0.030)  |
| $A^u$                            | −0.246***<br>(0.031) | −0.235***<br>(0.031) | −0.236***<br>(0.031) | −0.262***<br>(0.049)  |
| $A^d$                            | −0.396***<br>(0.032) | −0.381***<br>(0.032) | −0.417***<br>(0.052) | −0.418***<br>(0.052)  |
| $A^p$                            | 0.334***<br>(0.078)  | 0.407***<br>(0.079)  | 0.409***<br>(0.079)  | 0.417***<br>(0.095)   |
| <i>Endogenous regressors</i>     | none                 | AI                   | AI, DownAI           | AI and all three lags |
| <i>Kleibergen-Paap rk Wald F</i> |                      | 6591.3               | 3379.2               | 1783.9                |
| <i>Hansen J (p)</i>              |                      | 0.230                | 0.262                | 0.581                 |
| <i>Wu-Hausman (p)</i>            |                      | 0.0000               | 0.0000               | 0.0000                |
| <i>Controls</i>                  | Yes                  | Yes                  | Yes                  | Yes                   |
| <i>Firm FE</i>                   | Yes                  | Yes                  | Yes                  | Yes                   |
| <i>Year FE</i>                   | Yes                  | Yes                  | Yes                  | Yes                   |
| <i>Observations</i>              | 26,115               | 26,115               | 26,115               | 26,115                |

Note: The dependent variable is the youth employment share. Column (1) is least squares; columns (2) to (4) instrument the adoption terms named in the row “Endogenous regressors” with shift-share terms built from pre-sample occupational exposure and with the second- and third-order network lags of adoption. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

#### 4.7. From Firms to Cities: Reallocation or Destruction?

If the positive local coefficient in Table 5 is reallocation, then the same regression estimated on city aggregates must find a smaller number than the firm-level regression, because within a city the losses of the adopters and the gains of the non-adopters partly cancel. Table 8 estimates the same specification at three levels of the system.

At the firm level the coefficient is −0.434. At the city level it is −0.222 (0.095,  $p = 0.019$ ) — still negative, still significant, but only 51% as large. The implied reallocation share is 49%: about half of what the firm-level regression records as youth employment destroyed is youth employment that moved to another employer in the same city. At the industry level the coefficient is −0.493, close to the firm-level estimate, which is what one expects if reallocation is geographic rather than sectoral — displaced graduates find work in the same city, not in the same industry. The city-level coefficient for the experienced share is −0.056 and insignificant ( $p = 0.514$ ), so the local market is reallocating young workers specifically.

The last rows of Table 8 put the two halves of the argument side by side. The system-level total impact is −1.313, which is 3.02 times the firm-level estimate; the city-level estimate is about half of it. The firm-level coefficient is therefore not a lower bound on the aggregate effect, as a reader who thinks only about spillovers might assume, and not an upper bound either, as a reader who thinks only about reallocation might assume. It is a different quantity from both, and which direction it errs in depends entirely on the boundary the question draws around the system. A municipal government asking what generative AI is doing to its graduates should use something near −0.222; a ministry asking what it is doing to the cohort should use something near −1.313.



**Table 8.** The same coefficient at three levels of the system.

| Level of aggregation                         | Unit          | A         | SE      | N      | Within R <sup>2</sup> |
|----------------------------------------------|---------------|-----------|---------|--------|-----------------------|
| Firm                                         | firm-year     | −0.434*** | (0.018) | 26,115 | 0.097                 |
| Firm, network lags included                  | firm-year     | −0.422*** | (0.018) | 26,115 | 0.107                 |
| City                                         | city-year     | −0.222**  | (0.095) | 600    | 0.110                 |
| Industry                                     | industry-year | −0.493*** | (0.055) | 200    | 0.455                 |
| City, experienced share                      | city-year     | −0.056    | (0.086) | 600    | 0.106                 |
| Implied local reallocation share (%)         |               | 48.9      |         |        |                       |
| System-level total impact                    |               | −1.313    | (0.195) |        |                       |
| Ratio of total impact to firm-level estimate |               | 3.02      |         |        |                       |

Note: Each row estimates Equation (5) on the unit named in the second column; the city and industry panels are employment-weighted aggregates with unit and year effects. The implied local reallocation share is one minus the ratio of the city-level to the firm-level coefficient. The system-level total impact is taken from column (2) of Table 6. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

4.8. Is It Really the Network?

A spillover estimated from a network is only as good as the argument that the network, and not something correlated with it, is doing the work. Table 9 reports two partitions of the customer links that a network account predicts and its competitors do not.

Column (1) splits customers by input specificity. Adoption by customers buying relationship-specific inputs lowers the firm's youth share by 0.530 (0.044); adoption by customers buying generic, reference-priced inputs by 0.260 (0.045). The difference is −0.270 (0.063,  $p < 0.001$ ). This is the signature the propagation literature has documented for earthquakes and bankruptcies [17,19]: a shock travels along links the receiving firm cannot quickly replace. A common sectoral shock has no reason to respect input specificity.

Column (2) splits the same customers by geography. Adoption by customers in the firm's own city lowers its youth share by 0.317 (0.146); adoption by customers elsewhere by 0.400 (0.033). The difference is 0.083 (0.151,  $p = 0.584$ ) — statistically indistinguishable from zero. The shock travels along the supply chain and is indifferent to distance, which is precisely what distinguishes it from a local demand shock. Note that this is not in tension with the local reallocation result: the labour market is local, the supply chain is not, and the two channels are separately identified because they use different weight matrices.

Column (3) repeats the specificity split for the experienced share and finds a much smaller difference, and Table 10 makes the age comparison systematic. Every channel is larger for the young than for the experienced: own adoption by a factor of 5.05, suppliers' by 4.65, customers' by 4.14 and local peers' by 6.51, and every cross-equation test rejects equality. Hypothesis 5 is supported. The system does not merely transmit the shock; it transmits it to the same margin at every node.

**Table 9.** Which links carry the shock: input specificity against geographic proximity.

| Variable                         | (1)                  | (2)                  | (3)                  |
|----------------------------------|----------------------|----------------------|----------------------|
| <i>A</i>                         | −0.426***<br>(0.018) | −0.422***<br>(0.018) | −0.083***<br>(0.016) |
| <i>A<sup>u</sup></i>             | −0.249***<br>(0.031) | −0.246***<br>(0.031) | −0.052*<br>(0.028)   |
| <i>A<sup>d</sup>(specific)</i>   | −0.530***<br>(0.044) |                      | −0.069*<br>(0.038)   |
| <i>A<sup>d</sup>(generic)</i>    | −0.260***<br>(0.045) |                      | −0.122***<br>(0.041) |
| <i>A<sup>d</sup>(same city)</i>  |                      | −0.317**<br>(0.146)  |                      |
| <i>A<sup>d</sup>(other city)</i> |                      | −0.400***<br>(0.033) |                      |
| <i>A<sup>p</sup></i>             | 0.329***<br>(0.078)  | 0.331***<br>(0.079)  | 0.053<br>(0.070)     |
| <i>Difference tested</i>         | specific - generic   | near - far           | specific - generic   |
| <i>Difference</i>                | −0.270***<br>(0.063) | 0.083<br>(0.151)     |                      |
| <i>Controls</i>                  | Yes                  | Yes                  | Yes                  |
| <i>Firm FE</i>                   | Yes                  | Yes                  | Yes                  |
| <i>Year FE</i>                   | Yes                  | Yes                  | Yes                  |
| <i>Observations</i>              | 26,115               | 26,115               | 26,115               |
| <i>Within R<sup>2</sup></i>      | 0.107                | 0.107                | 0.051                |

Note: The dependent variable is the youth employment share in columns (1) and (2) and the experienced share in column (3). Each partition of the customer weights adds back to the full customer lag exactly, so the two components can be entered together. The difference is a Wald test that the two component coefficients are equal. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

**Table 10.** Age specificity of every channel.

| Channel               | Youth share          | Experienced share    | Difference           | Ratio |
|-----------------------|----------------------|----------------------|----------------------|-------|
| Own adoption          | −0.422***<br>(0.018) | −0.084***<br>(0.016) | 0.338***<br>(0.023)  | 5.05  |
| Suppliers' adoption   | −0.246***<br>(0.031) | −0.053*<br>(0.028)   | 0.193***<br>(0.041)  | 4.65  |
| Customers' adoption   | −0.396***<br>(0.032) | −0.096***<br>(0.028) | 0.300***<br>(0.041)  | 4.14  |
| Local peers' adoption | 0.334***<br>(0.078)  | 0.051<br>(0.070)     | −0.283***<br>(0.106) | 6.51  |

Note: Each row reports the coefficient on the channel named in the first column from Equation (6) estimated on the youth share and on the experienced share. The difference is the coefficient on the interaction of that channel with an equation indicator in the stacked model, whose covariance is clustered on the firm across both equations; the ratio is the youth coefficient divided by the experienced one. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

## 5. Further Investigations

### 5.1. Robustness

Table 11 reports 12 alternative specifications, showing the own-adoption and the customer-lag coefficients side by side. Replacing the year effects with industry-by-year and then city-by-year effects — the latter absorbing any shock common to a city in a year, including any local demand shock that might

masquerade as a network effect — leaves both coefficients essentially unchanged. Clustering on both the firm and the year, weighting by employment, restricting to the balanced panel, dropping the pandemic years, dropping state-owned firms, dropping the four largest cities, using the logarithm of the youth head count and adding the lagged dependent variable all leave the conclusion intact. Across the specifications measured in the units of the baseline the own coefficient ranges from  $-0.445$  to  $-0.371$  and the customer lag from  $-0.432$  to  $-0.370$ .

Two further checks address inference and omitted variables. A restricted wild cluster bootstrap with Rademacher weights and 999 replications gives  $p = 0.001$  for the customer lag [62,63]. The coefficient-stability bound of Oster [64] moves the customer-lag coefficient from  $-0.403$  without controls to  $-0.396$  with them while the R-squared rises from 0.037 to 0.107; the bias-adjusted value is  $-0.393$  and selection on unobservables would have to be 124 times as important as selection on the observed controls to drive the estimate to zero.

The sharpest check is in Appendix A. Table A1 replaces the customer weights with unweighted links and with the single largest customer, shifts the customer lag two years back and two years forward, and finally rewires the customer links at random while leaving every firm's own adoption, own characteristics and own outcome untouched. The lead specification is a placebo — a firm's youth employment today should not respond to its customers' adoption two years hence — and the rewired specification is the strongest available test that the network is the conduit. We also re-estimate the whole model on 300 random rewirings of all three matrices, preserving the degree distribution: the actual customer-lag coefficient is  $-0.396$  against a rewired distribution centred on  $-0.006$  with a standard deviation of 0.030, giving a randomisation p-value of 0.003; the supplier and local lags give the same answer. Whatever the estimated spillovers are, they are not a property of the firms — they are a property of who is connected to whom.

**Table 11.** Robustness checks.

| Specification                              | A              | SE      | A <sup>d</sup> | SE      | N      |
|--------------------------------------------|----------------|---------|----------------|---------|--------|
| Baseline                                   | $-0.422^{***}$ | (0.018) | $-0.396^{***}$ | (0.032) | 26,115 |
| Industry x year effects                    | $-0.371^{***}$ | (0.021) | $-0.385^{***}$ | (0.033) | 26,115 |
| City x year effects                        | $-0.388^{***}$ | (0.130) | $-0.396^{***}$ | (0.033) | 26,115 |
| Two-way clustering (firm and year)         | $-0.422^{***}$ | (0.023) | $-0.396^{***}$ | (0.025) | 26,115 |
| Employment-weighted                        | $-0.409^{***}$ | (0.028) | $-0.424^{***}$ | (0.048) | 26,115 |
| Balanced panel                             | $-0.422^{***}$ | (0.019) | $-0.370^{***}$ | (0.035) | 21,590 |
| Excluding the pandemic years               | $-0.445^{***}$ | (0.020) | $-0.396^{***}$ | (0.037) | 20,599 |
| Excluding state-owned firms                | $-0.413^{***}$ | (0.021) | $-0.432^{***}$ | (0.039) | 17,697 |
| Excluding the four largest cities          | $-0.421^{***}$ | (0.018) | $-0.378^{***}$ | (0.033) | 23,983 |
| Dropping firms with fewer than five links  | $-0.422^{***}$ | (0.018) | $-0.396^{***}$ | (0.032) | 26,115 |
| Youth head count in logs (different units) | $-0.023^{***}$ | (0.001) | $-0.023^{***}$ | (0.002) | 26,115 |
| Adding the lagged dependent variable       | $-0.436^{***}$ | (0.019) | $-0.417^{***}$ | (0.035) | 23,315 |

Note: Each row is a separate estimation of Equation (6) with the modification named in the first column. The specification in logs uses the logarithm of the number of employees aged 30 or below and is therefore not in the units of the other rows. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

## 5.2. Heterogeneity

Table 12 splits the sample four ways and reports both the own effect and the customer effect. The customer channel is stronger for non-state firms (−0.432 against −0.324), which is consistent with state-owned firms adjusting employment less freely for any given demand shock, and stronger for technology-intensive firms (−0.472 against −0.368), whose output is the kind of service that a customer's models can most readily replace.

Two splits speak directly to the systems argument. Firms that are central in the network — those whose supply-chain exposure moves most with the system — show a customer effect of −0.394 against −0.372 for peripheral firms: position in the system, not any characteristic of the firm itself, governs how much of the shock reaches it. And in thick local labour markets the own effect is attenuated relative to thin ones (−0.401 against −0.445), which is the reallocation result of Section 4.7 appearing again as a sample split: where there are many other employers, displaced graduates are absorbed.

**Table 12.** Heterogeneity analysis.

| Partition              | Subsample A     | A         | A <sup>d</sup> | Subsample B | A         | A <sup>d</sup> | Difference in A <sup>d</sup> |
|------------------------|-----------------|-----------|----------------|-------------|-----------|----------------|------------------------------|
| State ownership        | State-owned     | −0.441*** | −0.324***      | Non-state   | −0.413*** | −0.432***      | 0.109                        |
|                        |                 | (0.032)   | (0.055)        |             | (0.021)   | (0.039)        | (0.067)                      |
| Technology intensity   | High technology | −0.425*** | −0.472***      | Other       | −0.419*** | −0.368***      | −0.104                       |
|                        |                 | (0.033)   | (0.061)        |             | (0.021)   | (0.038)        | (0.072)                      |
| Network position       | Central         | −0.412*** | −0.394***      | Peripheral  | −0.430*** | −0.372***      | −0.022                       |
|                        |                 | (0.025)   | (0.043)        |             | (0.025)   | (0.053)        | (0.068)                      |
| Local market thickness | Thick           | −0.401*** | −0.448***      | Thin        | −0.445*** | −0.334***      | −0.114*                      |
|                        |                 | (0.024)   | (0.043)        |             | (0.026)   | (0.048)        | (0.064)                      |

Note: Each partition re-estimates Equation (6) on the two subsamples. Network centrality is the within-firm standard deviation of the supply-chain lag, split at the median; local market thickness is the number of listed firms in the city, split at the median. The difference is the customer-lag coefficient in subsample A minus that in subsample B, with a standard error computed from the two independent estimates. Standard errors clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and 10% levels.

**6. Discussion**

The claim of this paper is that the effect of generative artificial intelligence on youth employment is a property of a system of firms and cannot be recovered from any of them individually. The evidence is not one coefficient but an arithmetic. The within-firm estimate is −0.434. Add the supply chain and the total impact on the system is −1.313, 3.02 times larger, with 68% of it falling on firms that have adopted nothing. Restrict attention to a city and the estimate is −0.222, half the firm-level number, because within a city young workers move between employers rather than out of employment. The same underlying phenomenon supports three different numbers, and the only thing that determines which is right is where the boundary of the system is drawn.

This reframes a debate that has been conducted as though it had a single answer. Studies that find large effects and studies that find none may both be correct about their own estimand: a within-firm design in a thick local labour market will find reallocation and read it as a small effect, while the same technology is removing entry-level positions from the system at three times that rate through customers who never appear in the sample. The literature's disagreement is, at least in part, a disagreement about units that has been conducted as a disagreement about facts.

For firms, the practical implication is that exposure to the technology is not the same as adoption of it. A firm that has decided not to adopt has not thereby decided to be unaffected: if its customers adopt, its demand for junior labour falls anyway, and our estimates say that this second-hand exposure is larger than the first-hand kind. Workforce planning that looks only at the firm's own technology roadmap will therefore be wrong in a predictable direction, and the firms most exposed are those selling relationship-specific, junior-labour-intensive services to a small number of customers — precisely the firms least likely to think of themselves as being in the path of artificial intelligence.

For policy, three things follow. The first is a measurement warning: monitoring youth employment at the city level, which is how labour market statistics are collected, will systematically understate what is happening, because the local market is where reallocation is most complete. The second is that the instruments have to match the level. Local measures — placement services, wage subsidies, thicker local markets — genuinely help, and our heterogeneity results show they are already working: thick cities absorb displaced graduates. But they redistribute the burden rather than reduce it, and they cannot touch the supply-chain channel at all, because the customer whose adoption is closing a firm's entry port is typically in another city and another sector. The third is that the natural policy unit is the network. If propagation runs along relationship-specific links, then sectoral roadmaps, standards for AI-enabled procurement and transition support targeted at supplier ecosystems reach the affected firms in a way that firm-level instruments do not [65,66]. Recent systems-level work on Chinese firms points the same way, finding that the returns to artificial intelligence depend on the organisational, innovation and governance capabilities that surround it [67–70].

Three limitations bound the claims. First, the network is the disclosed one, which covers the five largest partners on each side; it is the economically important part of a firm's trade but not all of it, and measurement error in the weights will attenuate the network lags, making our multiplier a lower bound. Second, the sample is listed firms, which are the nodes of the network but not all of it; the unlisted suppliers and customers through which some of the propagation runs are invisible here, again in the direction of understatement. Third, the design identifies the effect over the four years after the shock. Whether the reallocation we document is temporary — the local market clearing while the system adjusts — or the beginning of a persistent reduction in entry positions cannot be settled with four years of data, and the answer matters, because a cohort that is reallocated is in a different position from a cohort that is displaced [34–36]. Following the same network as the technology matures is the obvious next study, and the organisational literature on how firms rebuild work around a new technology suggests what to look for: absorptive capacity governs what a firm can do with a new technology [71,72], the balance between exploiting existing routines and exploring new ones governs how fast it rebuilds them [73], and the way junior work is reorganised around the technology determines whether the entry-level rung is rebuilt or removed [74–78].

## 7. Conclusions

741 This paper asked what generative artificial intelligence is doing to youth  
742 employment, and answered that the question has no firm-level answer. Using  
743 26,115 firm-year observations for 2,800 Chinese A-share listed firms over  
744 2016–2025, their disclosed supplier and customer links and the 60 local  
745 labour markets they share, we estimated a network autoregressive model of  
746 the youth employment share and decomposed its impacts.

747 Own adoption lowers the youth share by 0.434 percentage points per  
748 standard deviation. Customers' adoption lowers it by a further 0.396 and  
749 suppliers' by 0.246, while local peers' adoption raises it by 0.334. The direct  
750 impact is −0.420, the indirect impact −0.893 and the total impact −1.313, a  
751 network multiplier of 3.12 with 68% of the effect outside the adopting firm; the  
752 city-level coefficient is −0.222, implying that 49% of the firm-level effect is  
753 local reallocation rather than destruction. Propagation is concentrated in  
754 relationship-specific links, is indifferent to geographic distance, is 4.1 times  
755 larger for young workers than for experienced ones, and vanishes when the  
756 network is randomly rewired.

757 The methodological conclusion is that the within-firm coefficient, which is  
758 what the literature reports, is neither an upper nor a lower bound on anything  
759 policy cares about. The substantive conclusion is that a young person's  
760 chances are being changed by technology decisions taken in firms they will  
761 never apply to. And the systems conclusion is the one in the title: for this  
762 question, the sum is not the parts, and the difference is large enough to  
763 reverse the reading of the existing evidence.

764 **Appendix A**

765 Table A1 reports alternative constructions of the customer network and the  
766 placebo tests described in Section 5.1. Replacing the disclosed sales shares  
767 with unweighted links, and with the single largest customer only, leaves the  
768 coefficient close to the baseline, so the result is not an artefact of the  
769 weighting. Shifting the customer lag two years back leaves it negative and  
770 significant, which is what a propagating shock should do. Shifting it two years  
771 forward is a placebo, since a firm's youth employment today cannot respond to  
772 its customers' future adoption. Rewiring the customer links at random, while  
773 leaving every firm's own adoption and characteristics exactly as they are, is  
774 the strongest test available that the estimated spillover belongs to the  
775 network rather than to the firms it connects.

776 **Table A1.** Alternative network constructions and placebo tests.

| Specification                            | Coefficient on                    | Estimate  | SE      | N      |
|------------------------------------------|-----------------------------------|-----------|---------|--------|
| Baseline customer lag                    | A <sup>d</sup>                    | −0.396*** | (0.032) | 26,115 |
| Unweighted (binary) customer links       | A <sup>d</sup> (unweighted)       | −0.387*** | (0.040) | 26,115 |
| Largest customer only                    | A <sup>d</sup> (largest customer) | −0.176*** | (0.017) | 26,115 |
| Customer lag two years earlier           | A <sup>d</sup> (t-2)              | −0.023    | (0.041) | 20,516 |
| Placebo: customer lag led by two years   | A <sup>d</sup> (t+2)              | −0.041    | (0.036) | 20,516 |
| Placebo: randomly rewired customer links | A <sup>d</sup> (rewired)          | −0.010    | (0.032) | 26,115 |

777 Note: Each row re-estimates Equation (6) with the customer lag replaced by the construction  
778 named in the first column; the coefficient reported is that on the replacement. The lead  
779 specification is a placebo, and the rewired specification reassigns customer links at random while  
780 leaving every firm's own adoption, characteristics and outcome unchanged. Standard errors  
781 clustered at the firm level are in parentheses. \*\*\*, \*\* and \* denote significance at the 1%, 5% and  
782 10% levels.

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